remembering titanic

Cross-curricular Lessons for Key Stage 3

About the Project

Many of the issues facing the world today are complex in nature and require knowledge and skills that cut across school subjects. The ability to analyse real-world issues and phenomena through different disciplinary lenses and to appreciate their interconnectedness has become an essential element of democratic citizenship.

Remembering Titanic was a collaborative project between researchers at the University of Southampton, the SeaCity Museum and seven teachers from state-funded secondary schools in the Southampton and Hampshire area. One of the key project priorities was to facilitate cross-curricular learning between different school subjects, namely science and history. As part of the project, we designed and delivered three half-day CPD sessions where the teachers learned about the Titanic disaster and developed an interdisciplinary curriculum unit. Below I describe the key ideas that underpinned the CPD and the lessons learned from the project.

Principles

Science & history learning	School-based and out-of-school learning
How are science and history similar or different as academic disciplines and school subjects? Where do the aims of science and history learning overlap in the curriculum? How can the two subjects work together to facilitate learning about the Titanic?	How can local resources and learning environments such as museums, monuments and other forms of memorials be utilised for learning about Titanic? How can school-based and out-of-school learning be integrated to facilitate meaningful learning?
Place-based learning	**Disaster and tragedy**
How can we engage students in learning about historical events that were significant to their own community?	How can we talk about tragic and disastrous events in a sensitive and respectful way?

People

The unit of work was a collaborative endeavour between the University of Southampton, science and history teachers from across Southampton and Hampshire and Museum Educators from SeaCity Museum, Southampton.

University of Southampton

Wonyong Park, Neta Shaby, Rachele Newman, Tilly Scotthorne

SeaCity Museum

Andy Skinner, Liza Morgan

Our teachers

Megan Barton, Christopher Dale, Ashleigh Fair, Natalie Harrison, Mary Shepherd, Catherine King, Christopher Zieba

Who It Is for

This unit of work can be used to develop pupil's key substantive and disciplinary knowledge of science and history at Key Stage 3. The story of the Titanic disaster has enduring appeal and interest for what it reveals to us about maritime design and development as well as social and economic history and the lessons that can be learnt when a disaster such as this occurs. It may be of particular interest to schools in the Southampton and Hampshire area but is likely to have a wider appeal due to the national and international interest in the sinking of Titanic.

How to Use

There are six lessons in this Unit of work. It was designed by science and history teachers to explore the Titanic disaster from a cross curricular perspective. The taught sequence has an element of flex in recognition that there may be timetabling or curricular constraints that make it difficult to align science and history taught content.

The four history lessons are designed to be taught in the identified sequence and they utilise previously taught content:

- **Lesson 1:** Why was Titanic significant to Southampton?
- **Lesson 2:** Why was the sinking of Titanic a disaster?
- **Lesson 3:** Who was most to blame for the Titanic disaster?
- **Lesson 4:** How far did the Titanic disaster have an impact on the world?

The two science lessons can either be slotted into the above sequence after Lesson 2 (Lesson A) and Lesson 3 (Lesson B) or at the end of the Unit of work.

- **Lesson A:** Watertight bulkheads
- **Lesson B:** Keeping afloat

Each lesson pack contains:

- **Lesson outline:** This outlines the intended learning, alignment to the National Curriculum and a brief narrative of how the lesson is designed to run.
- **Lesson presentation:** These PowerPoint slides will include all the main teaching points and activities in the lesson.
- **Lesson resources:** Any source material or resources used in the lesson will be included here.

Curriculum Goals

The unit of work is intended to meet several of the National Curriculum goals for both Science and History. These include:

Science

- Practical enquiry into the nature, processes and methods of science through different types of science enquiries that help them to answer scientific questions about the world around them
- Describe associated processes and key characteristics in common language, and… technical terminology accurately and precisely
- Ask questions and develop a line of enquiry based on observations of the real world as well as making their own predictions using scientific knowledge and understanding.
- Select, plan and carry out the most appropriate types of scientific enquiries to test predictions
- Use appropriate techniques, apparatus, and materials during laboratory work, paying attention to health and safety

History

- Understanding historical concepts, including historical significance
- Gain historical perspectives by placing knowledge into different contexts
- Understand the connections between local, regional, national and international history, between cultural, economic, military, political, religious and social history; and between short- and long-term timescales
- Study of an aspect or site in local history dating from a period before 1066
- Pursue historically valid enquiries
- Understand how different types of historical sources are used rigorously to make historical claims and discern how and why contrasting arguments and interpretations of the past have been constructed
- Make connections and draw contrasts
- Identify significant events, make connections, draw contrasts (focusing on the concepts of change and continuity)

Project History

Workshop 1	February 2023 SeaCity Museum	Introductions Purpose of the project Why integrate science and history? Benefits and approaches Comparing the Key Stage 3 Science and History programmes of study Why is informal education important? Strategies for integrating informal and formal learning Guided tour of the SeaCity Museum Sharing initial ideas about the unit
Workshop 2	May 2023 University of Southampton (Boldrewood Campus)	Place and community-based learning Disaster and tragedy education Developing a skeletal scheme of work for six lessons Planning workflow Wrap-up
Workshop 3	June 2023 University of Southampton (Avenue Campus)	Developing lesson resources Exchanging feedback Reflections on the project
Remote collaboration	July to September 2023	Collating teacher and student materials

Additional Information

Remembering Titanic: https://www.remembertitanic.com/
SeaCity Museum: https://seacitymuseum.co.uk/

Contact

Wonyong Park (w.park@soton.ac.uk)

Lesson 1: Why was the Titanic significant to Southampton?

Summary of Lesson

Objectives:

- To make inferences from individual experiences about why the Titanic was significant to Southampton.
- To reach a well explained judgement on why the Titanic was significant to Southampton.

This is the first lesson in the scheme of work and focuses on why the Titanic mattered to Southampton at the time of its launch in 1912. Pupils will consider this by applying significance criteria to the information they gain and infer from the experiences of individuals connected to the Titanic. This lesson uses the 5 Rs significance criteria, but this can be changed to fit in with what departments and students are used to working with. The lesson begins with a brief overview of what the Titanic was the historical context. Pupils will then engage with an interactive map of Southampton, showing where passengers and crew were from in the city and whether they survived the disaster. From this, pupils can begin to discuss why the Titanic was significant. From here, pupils look at six individuals and apply the significance criteria to each experience. Students should then be able to draw on their knowledge from the lesson to reach an overall and explained judgement on why the Titanic was significant.

National Curriculum Links

This lesson links to the national curriculum aims of:

- 'understanding historical concepts such as…significance'
- 'gain historical perspective by placing their growing knowledge into different contexts, understanding the connections between local, regional, national and international history; between cultural, economic, military, political, religious and social history; and between short- and long-term timescales'.

The lesson links to the content that should be taught by being an example of 'challenges for Britain, Europe and the wider world 1901 to the present day'.

If teaching in Southampton or Hampshire, the lesson also links to the requirement of having a local history study.

Resources Required

- Lesson PowerPoint
- Significance table
- Individual experiences information

Lesson Guidance

Step 1: As this is the first lesson in the scheme of work, the lesson begins with what pupils already know about the Titanic. This is chance to gauge what knowledge pupils already have and address any misconceptions. Pupils complete a brain dump task, where they write down everything they already know. This could be done as a mind map or list in books, or pupils could write their ideas on post its and bring them to the board. There is summary video linked give students an overview, after their ideas have been shared and discussed.

Step 2: This lesson is built around the concept of significance. At this point, the criteria pupils will be working with is shared. Start with some recall of what pupils know significance to mean and where/when they have seen/applied it before in history. This lesson uses Counsell's (2005) 5Rs criteria for Historical Significance. Explain what each of these elements means to students and that they will be applying: revealed, resulted in changed and remarkable in the lesson. This lesson will work with different significance criteria so please do adapt the slides and resources should you wish to use an alternative to the 5Rs. For a higher ability group, all 5Rs could be used or the class could generate their own criteria.

Step 3: Now pupils are familiar with what the Titanic was and the significance criteria they are using, the lesson now focuses on Southampton at the time of the Titanic's launch. Slide 6 contains some brief context on Southampton in the early 20th Century. Pupils could be questioned at this point on the following:

- What does this reveal about Southampton at the time?
- Why was Southampton chosen as the port Titanic was to launch from?
- What would the launch mean for Southampton? (In terms of status, economy, opportunities for residents).

Step 4: Building on the initial context, pupils will now look at a source: an interactive map of Southampton showing where the people from the city on board the Titanic lived and what happened to them. If teaching in the city, you could look at the area that the school is in or where pupils live. This task should start to get pupils thinking about why the Titanic was significant to Southampton. Pupils could consider the following: Which parts of the city were most of the crew from? Why could this be? What would this mean for these areas? What would the large losses of life mean for these areas?

Step 5: At this point in the lesson, pupils have a chance to reflect on the lesson question and share their thoughts so far. This could be done in variety of ways, for example mini whiteboards, think pair share, post-its.

Step 6: The lesson now moves onto pupils working on the main task independently. The task builds on the thinking pupils will have done up until this point and gets them to apply the criteria revealing, resulted in change and remarkable to six induvial experiences. Each pupil will need a copy each of the significance table. This can be amended to include further prompt questions should pupils need them. Pupils are to complete the table using the six profiles. This could be done in several ways. Pupils could be given an individual each, become an expert and then move around the room sharing their individual's experiences and recording what they learn from others. The information could be put around the room and pupils move around. Or pupils could work at desks from the sheet. Below is a list of example inferences pupil might make:

- **Smith** – Southampton had a reputation as a port city, many important people from the maritime industry lived/worked there. The Titanic provided opportunities for experienced sailors. The captain being a Southampton resident would have brought pride and status to the city. As captain, Smith would have been a well-known figure and discussed in Southampton following the disaster.
- **Boxhall** – Life could be tough for people in Southampton and the Titanic offered important work and a chance to gain experience and potentially status.
- **Fleet** – There were a variety of roles on the Titanic. As the man who spotted the iceberg, Fleet would have been remarkable both nationally and locally.
- **Evans/Scarrott** - Life could be tough for people in Southampton and the Titanic offered important work and a chance to gain experience and potentially status. The loss of lives would have brought grief to local communities and put pressure on their families due to a loss of income.
- **Joughin** – The port provided work for Southampton residents from a young age. Joughin experience and unlikely survival would have been remarkable both locally and nationally.

Step 7: Pupils should now be able to reach a judgement on why the Titanic was significant to Southampton. Slide 11 has a few prompts to encourage pupils to think hard. These could be altered or removed depending on the group. This activity could be run as a discussion task or a written one, with written ideas later shared with the class, for example under a visualiser.

Step 8: If time, pupils can consider and discuss the significance of Titanic to Southampton today using the various monuments around the city as a starting point.

Lesson 2: Why was the sinking of the Titanic a disaster?

Summary of Lesson

Objectives:

- To use historical evidence related to the sinking of Titanic to reach a causal conclusion
- To reach an evaluative judgement as to why Titanic is viewed as a historic disaster

This lesson looks at the actual sinking of the ship and why it then became an event in history that can be categorised as a disaster. The lesson is broken into 3 parts:

1. Basic breakdown of the ship
2. How it actually sank
3. Why it became a 'disaster'

The first two parts can be completed relatively quickly but and the main task is looking at why it is an event in history that can be categorised as 'disaster, 'with a breakdown of what this means as well as links to other events.

National Curriculum Links

This lesson proved the opportunity for students to:

- study of an aspect or site in local history dating from a period before 1066
- gain historical perspective by placing their growing knowledge into different contexts with a focus on the concept of a 'disaster'.

There are clear links for schools in Southampton and along the South Coast. This may also be useful for schools in regions which also have connections to Titanic such as Liverpool and Merseyside, Belfast, Cobh and Cork.

Resources Required

The resources for this lesson are optional but slides ca be taken and used at various points to support knowledge and understanding.

- Images (taken from the slides) for students to either have in their books to refer to.
- The task on slide 11 can be made into a table/other format if it is easier.

Lesson Guidance

The lesson is almost split into two parts. The first part looks at why the ship was considered to be 'unsinkable'. This is done through looking at her construction, design and a newspaper article from the time as well how this was countered by ship building company that built the ship. Slides 2, 3, 4 and 5 look into the different reasons for why people though that it was unsinkable and students can break each of these down. For example, on slide 2 it is important to note that there were' watertight compartments'. Questions can be asked around why newspapers have such an impact and so why the belief was widely held as well as well as questions around why it was so important for people to believe that it was unsinkable. Can also link to the industrial revolution if this has been previously studied. Slides here can be printed out and annotated if required.

The second part of the lesson looks at why it eventually did sink. The video on slide 6 explains how the ship sank, looking at and linking to the construction and design of the ship and well as some of the technology on board. Students could use this to write notes/questions what happened as the ship went down. It would be worth spending a bit of time on this as it breaks down well the sinking event itself.

Then students are going to begin to break down the definition of a disaster and look at all the ways the sinking of the Titanic became one. This could also be moved to closer to the start of the lesson depending on how key terms, words or language are taught. Examples of other disasters could be given to give students comparison to other events, such as a volcanic eruption.

Students are then going to watch another short video on all the reasons for why the ship was unprepared to deal with different events happening, which will lead them to think about why this became such a massive incident. This can be mind-mapped or bullet-pointed, whichever way the teacher prefers. Students could find links between different reasons as well ask think of any questions they might have for why these things weren't sorted and/or why it ended as it did.

Lastly, using the information they have collected as well as new information on the board, they are going to categorise their ideas into the following sections:

- Before departure

- During the collision
- After the collision

Things to consider here would be questions around:

1. Could this have been avoided?
2. If so, how?
3. How would people feel if they knew the Titanic was not prepared as well as it could have been?

On Slide 12, there is a quote from a passenger from the ship and this can be discussed if there is time.

Lesson 3: Who was to blame for the sinking of the Titanic?

Summary of Lesson

Objectives:

- To construct an interpretation as to who was most to blame for the Titanic disaster
- To compare and critically analyse British and American enquiries

Throughout this lesson, students will build on their prior learning from the previous lessons looking at why the Titanic disaster happened and why it was significant. Using the different cause factors identified in lesson 2, students will now apportion blame for these errors, building to them creating their own interpretation of "Who was to blame". Students will then have the chance to compare their interpretation to two real life views, that of the US and British enquiries into the disaster.

National Curriculum Links

This lesson links to the National Curriculum aim for pupils to:

- pursue historically valid enquiries
- understand how different types of historical sources are used rigorously to make historical claims and discern how and why contrasting arguments and interpretations of the past have been constructed
- make connections, draw contrasts" focusing on the concepts of differing interpretations.

If in the local area to Southampton or another Titanic site, this lesson provides the opportunity for students to study a local history study linking to the:

- challenges for Britain, Europe and the Wider World 1901 to the present day

Resources Required

- Lesson PowerPoint
- Resource 1: Worksheet on side 8

Lesson Guidance

Slide 1-2: Five question recall quiz for students to complete at the start of the lesson. This is in keeping with Rosenshine's Principals of instruction (2012) which suggests that regular daily review and checking for students understanding are both vital for new learning to take place. Answers on slide 2

Slide 3: Display Learning Objectives to students and explain the rationale for today's learning.

Slide 4: Introduce new vocabulary, "Culpable" and "Interpretation". We are going to be creating our own interpretation as to who was most to blame (or in other words, most culpable) for the sinking of the Titanic, as well as looking at different interpretations from the US and British enquiries.

Slide 5: Ask students to think back to the previous lessons. Which people have we met that might have played a part in causing the Titanic to either sink, or led to it being such a disaster? At this point, students might not recall specific names or people, but might be able to recall 'The Captain', 'The lookout', 'The Designer' etc.

Slide 6: Now to put some names to these positions. Introduce students to our 6 key people that we are going to consider as possibly being most culpable for the disaster. Firstly Captain Smith, should he take the blame as the ships captain? What about Thomas Andrews, the ships designer? There were several changes that could have been made to the design (such as waterproof bulkheads, stronger rivets and more lifeboats) which if he had incorporated these into his design, would have mitigated the disaster. Or perhaps the blame should lie with First Officer William Murdoch, who had been in charge at the time when it hit the Iceberg? Another often cited culprit is J Bruce Ismay, chairman of the White Star Line company. Was it his cost cutting measures, and demand for increased speed that led to the disaster? Everyone knows the Titanic sank because it hit an iceberg, so surely some of the blame should be given to Frederick Fleet, the lookout who failed to spot it? And finally, what about the culpability of Stanley Lord, Captain of the nearby California, who ignored the distress calls of the Titanic? If he had intervened, the death toll would likely be dramatically reduced.

Slide 7: The embedded video clip is a song from 'Titanic: The Musical', sung by Andrews, Smith and Ismay. In it, they all blame each other for the disaster. Ask students to listen through to the song and read the accompanying lyrics. What reasons do they each put forward for blaming the others?

Slide 8: Worksheet. Print for students. Students to read the information in the first column. In the second column, ask them to fill in who this suggests was to blame, as well as giving an explanation as to why. In the final column, pupils can reflect on if there is anything that could have been done differently to reduce the severity of the disaster.

Slide 9: Ask students to create their own interpretation of who was most culpable for the disaster, as well as explaining why they have taken this view.

Slide 10-11: Following the disaster, there were enquiries carried out by both the British and the Americans. On these slides you will see summaries of their findings. Ask pupils who each enquiry blames for the disaster, do they agree?

Slide 12: Consolidatory writing activity, pupils to answer the 3 questions. This can then be fed back as a class.

Lesson 4: How far did the Titanic disaster have an impact on the world?

Summary of Lesson

Objectives:

1. To make inferences from historical sources about the local and national impact of the Titanic disaster.
2. To analyse the impact on sailing and shipping by identifying changes over time.

This lesson will be focusing on the impact the Titanic disaster had on a local, national and international level, utilising the concepts of change and continuity and also source analysis. The lesson begins by looking at the impact it had on the city of Southampton, as well as including national links such as charity fundraising that the people of the UK supported. From this, students will predict what they think would change after the sinking of the Titanic thinking about its impact. To test these predictions, students will analyse the change and continuity to shipping by looking at legal developments regarding safety measures as well as looking at other disasters. Students should be able to evaluate that although there was considerable change to shipping on an international level, disasters still occur due to human error, unexpected conditions and naturally occurring faults.

National Curriculum Links

This lesson links to the National Curriculum aims of:

- gain historical perspective by placing their growing knowledge into different contexts, understanding the connections between local, regional, national and international history
- pupils to understand the process of change
- identify significant events, make connections, draw contrasts (focusing on the concepts of change and continuity).

If in the local area to Southampton or another Titanic site, this lesson provides the opportunity for students to study a local history study linking to the:

- challenges for Britain, Europe and the Wider World 1901 to the present day.

Resources Required

- Lesson PowerPoint
- Resource 1: Images 1-6
- Resource 2: Source worksheet
- Resource 3: Information cards
- Resource 4: Line of change and continuity

Lesson Guidance

Step 1: There is an image of a cheque made out to the Titanic Relief Fund for £1 and 10s. Ask students to make some inferences from the source about the impact of the Titanic being aware that some students may not know what a cheque is. Use probing questions to draw out ideas. Reveal the provenance of the source: a cheque donated to the Titanic Relief Fund, a national charity set up following the disaster to support the families and victims of the sinking. The donation of £1 and 10s, is around the equivalent of £120 so a sizeable donation has been made. Here you can further probe the students about what this reveals about the impact of the Titanic raising ideas about people's feelings towards supporting the relief fund, you could draw comparisons here with charities that people donate to and why (Children in Need, Race for Life etc).

Step 2: This task is to make inferences from historical sources about the local and national impact of the Titanic disaster.

Explain to the students that they are going to use images from the Sea City Museum in Southampton, to explore how Southampton and the UK reacted to the disaster to identify the impact it had at the time. Questions to use are below:

1. Does the source show Immediate/short term/long term impact?
2. Local or national

3. What can we learn about the impact it had on Southampton and the UK?

Use Source 1 as a modelled example. Students can then work independently looking at the other sources and grasping an understanding of the impact that the Titanic had on both the local and national level.

After the source analysis, question students about their findings from the sources. What have they revealed about the impact of the Titanic? What might be the consequences of these? These questions may also be useful to use during the source task to challenge students and/or prompt their thinking.

Example questions:

- The date on the cheque is 1917, what does that reveal to us?
- If there were husbands lost, what did this mean for the families left behind?
- What does the clothing of the people in source 3 reveal to us about the impact and expectations of news?
- A football match had its date moved - why might they have moved the date?

Students should get the idea that the main impact of the Titanic was the loss of life in the city and that there was huge national support for raising money and to memorialise those that were lost.

Step 3: Following their analysis of the impact of the disaster, students are going to look at what the enquiries suggested should change to avoid another disaster. To begin this analysis, ask students: With the impact of the Titanic stretching across the UK, what changes do you think there would be after the disaster?

Following their ideas show them the recommendations that were made. These are a brief summary for students but further information can be found at https://www.titanicinquiry.org/

Using the recommendations identified, students should analyse what changes have taken place since the Titanic disaster and what has continued. Provide students with a set of information cards. Students are going to use a line of change and continuity to look at how sailing and shipping has or has not changed over time. Ask them to read the information and place the information cards along the line according to the level of change and continuity.

When modelling the task, probe students with the following questions:

Example questions

- Does the event show change or continuity?
- If change - is it considerable change?
- Is it law or a recommendation?
- Is it national or international?
- How might this impact on the change it could have?
- Have things really changed over time- has anything stayed the same?
- What continuities can you find? Why are there still continuities with disasters? What themes come from the events that show there are continuities?

Students should be able to identify that safety measures and technology has changed since the disaster, and that laws to improve safety have been made. They should also be able to evaluate that this does not always work in practice and that disasters still happen for different reasons. After completion, pose the question: 'What does this new information tell us about the impact of the Titanic?'

Final step: This plenary is for students to finalise their evaluations on the change and continuity that the Titanic disaster has had on shipping. Students can do this in a number of ways depending on the time allocated.

1. Students could complete a formal piece of writing on their findings about the impact (locally, nationally or internationally) or their views on change and continuity producing an explanation on their view.
2. Students could explain their position with regards to how far they agree there was change. This could be done verbally, physically or written.
3. Students could be introduced to more philosophical questioning or discussions around disaster, such as "Why could the lessons learnt from the Titanic not prevent further disasters".

Lesson A: Watertight Bulkheads

Summary of Lesson

Objectives:

1. To be able to describe the function and operation of bulkheads
2. To investigate the effectiveness of bulkheads
3. To evaluate the effectiveness of bulkheads through experimentation.

This lesson introduces the concepts of bulkheads to students and how they work through practical experimentation. Students will be able to 'select, plan and carry out the most appropriate types of scientific enquiries to test predictions' and finally to 'use appropriate techniques, apparatus, and materials during laboratory work, paying attention to health and safety'. They will also be able to record their results and test come to their own conclusions at the end. This lesson can fit in, if not possible with the coinciding history lesson due to curriculum restraints, with a unit on forces or density (floating and sinking) and can be followed up using "Lesson B: Keeping Afloat".

National Curriculum Links

This lesson focuses on the knowledge and conceptual understanding through:

- practical enquiry into the nature, processes and methods of science through different types of science enquiries that help them to answer scientific questions about the world around them.

By focusing on the sinking of the Titanic this gives students the context of bulkheads in a historical event, but also allows them to explain the sinking of the Titanic by describing:

- associated processes and key characteristics in common language, and [using] technical terminology accurately and precisely.

Equipment

- Lesson PowerPoint
- A large cardboard drinks carton (e.g., 1 litre juice carton)
- Scissors
- Ruler
- Stopwatch
- Waterproof tape
- Two large plastic storage boxes (larger than 10 litre) filled with tap water (to simulate water tanks)
- Resealable waterproof bag
- White clay or play-dough

Lesson Guidance

Step 1: Ask students to write down what they know about the Titanic. If completing the Science lessons alongside the History lessons, please see the resources of these to identify what prior knowledge they should be able to recall. Collate ideas as a class regarding the sinking of the Titanic and question them as to why they think the Titanic sank.

Go through the background detail of the Titanic and explain the concepts of the Watertight Bulkheads and how they function. You may wish to give students a copy of the diagram to annotate with notes.

Step 2: The practical experiment

Ask students to:

1. Follow the instructions on the PowerPoint to build two identical model ships: one without a bulkhead, and one with a bulkhead.
2. Place equal weights (clay in resealable bag) inside both ships to simulate the cargo and passengers. Ensure the weight is evenly distributed.
3. Float the ships in their respective water tanks and observe if they remain stable and at the same water level.
4. Using a pencil, poke a small hole at the same height (about 1.5 cm from the bottom) on both ships to simulate a breach in the hull.
5. Observe and time how long it takes for each ship to begin to sink. Observe and record what happens in each ship. Repeat this process three times.

Step 3: Ask students to feedback their results and findings from the practical regarding the time it took with/without a bulkhead. Students can record these conclusions or discuss as a class. Explain what happened to the bulkheads on Titanic.

Teaching Notes

The ship with a bulkhead will flood more slowly (only the compartment with the hole) and not sink completely. This can be related to the purpose of bulkheads: to separate compartments, maintain buoyancy and control damage. If the observed result is not as expected, check the sealing of the bulkhead, the weight and balance of the clay and the size of the hole.







Lesson B: Keeping Afloat

Summary of Lesson

This lesson is aimed at coinciding with the correlating history lesson ('Why did the sinking of the Titanic become a disaster?'), although this is not essential. It is to follow on from a lesson on watertight bulkheads for students to put to the test their own theories on how they could keep their own 'ship' afloat.

National Curriculum Links

This lesson proved the opportunity for students to practice their experimental skills and investigations. Students will be able to:

- ask questions and develop a line of enquiry based on observations of the real world as well as making their own predictions using scientific knowledge and understanding.

This lesson also allows them to:

- select, plan and carry out the most appropriate types of scientific enquiries to test predictions
- use appropriate techniques, apparatus, and materials during laboratory work, paying attention to health and safety.

They will also be able to record their results and test come to their own conclusions at the end.

If it is not possible to align this lesson with the accompanying history content due to curriculum restraints, this lesson can be built into a unit on forces or density (floating and sinking).

Equipment

- Lesson PowerPoint
- A large cardboard drinks carton (e.g., 1 litre juice carton)
- Scissors
- Ruler
- Stopwatch
- Waterproof tape
- Two large plastic storage boxes (larger than 10 litre) filled with tap water (to simulate water tanks)
- Resealable waterproof bag
- White clay or play-dough

Lesson Guidance

This lesson is a practical lesson with the two aims of:

- to investigate different 'anti-sinking' ideas
- to evaluate the effectiveness of different 'anti-sinking' ideas

To being with, students are expected to complete a factual recall of the previous lesson in order to use that knowledge in today's lesson. This will not take long and can be completed either through written or verbal answers. To support this, on Slide 4 there is a diagram that can be explained if needed.

Once the aim on Slide 5 has been explained, students can begin to prepare for their practical by gathering their resources.

It is important to note the students will be expected to record their findings on the results table on Slide 7 whilst they are working so they can then write their conclusions are the end. This can finally be linked back to the Titanic right at the end where student can see if any of their ideas could have potentially helped the Titanic before it sank.

Teaching Notes

Students can manipulate the mass and distribution of clay (representing cargo and passengers) and the number and position of bulkheads to explore what variables can affect the flooding of a ship. They should be able to control the variables they are not testing.

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